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Artificial Intelligence in the New Era of Decision-Making: A Case Study of the Euro Stoxx 50

Javier Parra-Domínguez * and Laura Sanz-Martín

BISITE Research Group, University of Salamanca, 37007 Salamanca, Spain; laurasanzmartin@usal.es
* Correspondence: javierparra@usal.es

Abstract: This study evaluates machine learning models for stock market prediction in the European stock market EU50, with emphasis on the integration of key technical indicators. Advanced techniques, such as ANNs, CNNs and LSTMs, are applied to analyze a large EU50 dataset. Key indicators, such as the simple moving average (SMA), exponential moving average (EMA), moving average convergence/divergence (MACD), stochastic oscillator, relative strength index (RSI) and accumulation/distribution (A/D), were employed to improve the model's responsiveness to market trends and momentum shifts. The results show that CNN models can effectively capture localized price patterns, while LSTM models excel in identifying long-term dependencies, which is beneficial for understanding market volatility. ANN models provide reliable benchmark predictions. Among the models, CNN with RSI obtained the best results, with an RMSE of 0.0263, an MAE of 0.0186 and an R^2 of 0.9825, demonstrating high accuracy in price prediction. The integration of indicators such as SMA and EMA improves trend detection, while MACD and RSI increase the sensitivity to momentum, which is essential for identifying buy and sell signals. This research demonstrates the potential of machine learning models for refined stock prediction and informs data-driven investment strategies, with CNN and LSTM models being particularly well suited for dynamic price prediction.

Keywords: artificial intelligence; finance; prediction models; financial decision-making; neural networks

MSC: 68T07

Citation: Parra-Domínguez, J.; Sanz-Martín, L. Artificial Intelligence in the New Era of Decision-Making: A Case Study of the Euro Stoxx 50. *Mathematics* **2024**, *12*, 3918. https://doi.org/10.3390/math12243918

Academic Editors: Shuo Yu and Feng Xia

Received: 19 November 2024
Revised: 6 December 2024
Accepted: 11 December 2024
Published: 12 December 2024



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1. Introduction

Artificial intelligence (AI) is now a familiar concept, which is becoming a fundamental part of our everyday lives and has gained significant prominence across various sectors in recent years. However, the roots of AI extend much further back, with the first explorations of the subject emerging in the mid-20th century. British mathematician Alan Turing was a pioneer in this field, laying the groundwork for AI with two influential papers, "Computing Machinery and Intelligence" [1] and "Intelligent Machinery" [2], which outlined his vision for machines capable of replicating human intelligence.

While numerous definitions of AI exist, ref. [3] describe it as technology that enables machines, devices, and software applications to simulate human intelligence, learning from experience through iterative processes and algorithmic training. From a broader perspective, ref. [4] liken the rise of AI to the Industrial Revolution, where technological advancements transformed manual tasks and processes that had defined human limitations for centuries. These authors argue that AI holds similar revolutionary potential, with the ability to reshape and reimagine human tasks and activities across industrial, intellectual, and social domains.

Moreover, AI's evolving capability has unlocked new frontiers, from transforming industries to solving complex problems that were previously considered intractable. Its role in finance, healthcare, and autonomous systems highlights the immense possibilities for innovation and raises critical ethical questions about its limits and responsible use. As AI matures, understanding its impact across sectors becomes increasingly essential, guiding technological and policy-driven decisions.

AI plays a crucial role in the financial sector, benefiting a significant portion of the population. Its primary applications in finance include optimizing investment portfolios, predicting trends and prices of financial assets, and performing sentiment analysis on news and social media to assess public opinion regarding companies and assets.

As part of technological progress, AI has radically transformed the financial industry [5,6]. An in-depth exploration of the EU50 index is an excellent way to reveal our current importance and the range of possibilities and challenges we face. A common practice is the development of predictive models, either for future earnings or for bankruptcy [7,8]. Pioneering work in bankruptcy prediction, such as Altman's Z-score model [9], has laid the foundation for understanding corporate financial distress. His model and subsequent developments [10] remain influential in assessing a firm's likelihood of bankruptcy. Stock markets play a crucial role in any developing economy, and their prices are significant indicators of a country's economic growth [11]. Investment is primarily aimed at maximizing returns and minimizing risks. Financial markets are characterized by their dynamism, unpredictability, and non-linear nature, which makes predicting their prices or trends a complex challenge influenced by numerous factors [12]. In recent years, stock market prediction has become a complex problem to solve [13], and experts from various parts of the world are working hard to predict the values of stocks in the market as accurately as possible in order to make the best investment decisions. However, there is no consensus on the effectiveness of these models [11], and the perfect predictive tool will likely never be achieved. In this regard, a wide variety of parametric and non-parametric statistical methods have been employed to address this challenge [14]. Currently, advanced techniques based on technical or fundamental analysis are used to predict stock prices [12].

In addition to historical prices, today's stock market is influenced by the mood of society. Financial markets are based on expectations, and investors buy or sell stocks based on fear or desire caused by some event. They rely on the credibility of media outlets that publish rumors despite knowing that the information is not always reliable, and this trend is currently growing with the use of social networks such as X, Facebook, Instagram and YouTube [15]. Therefore, incorporating information from social networks and historical prices can enhance the predictive capacity of models used in this context [16].

This study conducts a comparative analysis of different neural network architectures and technical indicators to enhance decision-making in the European stock market, specifically focusing on the EURO STOXX 50 index. By employing advanced machine learning techniques, such as neural networks, our research aims to improve the accuracy of stock price trend predictions and provide actionable insights for financial decision-makers. In this way, this work evaluates the effectiveness of three prominent models—artificial neural networks (ANNs), convolutional neural networks (CNNs), and long short-term memory networks (LSTMs)—in predicting stock market trends. The analysis identifies the strengths and limitations of each model in a real-world context, offering practical guidance for investors, financial analysts, and algorithm developers. ANNs act as a reference point in this research, bridging the gap between traditional neural network techniques and more advanced architectures, while also showcasing their effectiveness when paired with appropriate features. Their inclusion highlights the practicality of simpler architectures in achieving competitive performance, particularly in scenarios where computational resources are limited. Additionally, ANNs provide a clear baseline for evaluating the improvements offered by more specialized models like CNNs and LSTMs, ensuring a comprehensive comparison of predictive capabilities. This study contributes to the

growing body of knowledge on artificial intelligence in financial markets by bridging technical model performance with practical decision-making in the European stock market.

This paper is organized as follows: we will start with a state-of-the-art discussion of the implications of artificial intelligence in finance. Then, in the Materials and Methods section, we will present the database, which comprises the EU50 index, and the methodology used. This methodology involves the use of neural networks such as Dow theory, technical indicators, evaluation metrics, and data partitioning strategy and software. In the case study, we will show the prediction model developed and the results obtained by applying artificial intelligence techniques to the European financial market. Finally, in the Conclusions, we will analyze the findings of our research, assessing their relevance and implications for practice and future research.

2. Literature Review

We have conducted a literature review to identify and analyze the main articles exploring the intersection between artificial intelligence and the areas of finance and accounting. The articles were selected from Scopus and Web of Science databases in light of their relevance to the topic and their citation count, ensuring that the most impactful and significant research is included in our analysis. This analysis aims to provide a clear picture of how artificial intelligence is being applied in these fields and what results and trends are emerging from current research.

Table 1 lists various studies exploring AI applications in finance by predicting financial crises, proposing federated learning frameworks for IoT, examining the impact of AI on financial inclusion, studying AI applications in credit assessment and portfolio management, optimizing a CNN for stock market prediction, developing a model for predicting stock prices or combining artificial neural networks with other methods for financial prediction.

Table 1. State of the art: principal articles.

Reference	Objective	Methodology	Main Findings
[17]	To develop a model for predicting financial crises and ranking firms based on their likelihood of crisis.	Financial Crisis Prediction (FCP) model using Ant Colony Optimization (ACO), tested on five datasets.	The ACO-based model outperformed other machine learning models, providing financial institutions with a tool for more accurate crisis predictions.
[18]	To improve collaborative model training without compromising data privacy in smart grids.	Federated learning in IoT smart grids, using over-the-air computing (AirComp) and reconfigurable intelligent surfaces (RISs).	The proposed approach reduced training loss and improved prediction accuracy compared to traditional methods.
[19]	To examine how AI impacts digital financial inclusion for underserved populations.	AI tools optimizing risk management, chatbots for customer support, and alternative credit scoring.	AI significantly improves financial inclusion by addressing information asymmetry and broadening access to credit.
[14]	To evaluate AI applications in credit assessment, portfolio management, and financial forecasting.	A comparative study of AI techniques in finance: Artificial Neural Networks (ANNs), Expert Systems (ESs), and Hybrid Intelligent Systems (HISs).	AI offers higher accuracy when it comes to non-linear financial problems, but more research is needed to optimize its use in finance.
[20]	To predict stock index movement using CNN and GA for temporal patterns in multivariate time series data.	A multi-channel Convolutional Neural Network (CNN) optimized with a Genetic Algorithm (GA) for stock market prediction.	The GA-CNN model outperformed ANN and CNN in terms of prediction accuracy and convergence speed, showing its potential in financial forecasting.

[21]	To predict stock prices (opening, lowest, and highest) using historical data and technical parameters.	Associated deep recurrent neural network (LSTM-based) for simultaneous stock price prediction.	The associated network model outperformed standard LSTM and DRNN models, achieving over 95% accuracy in predicting multiple stock prices.
[22]	To improve stock price prediction by optimizing ANN weights with GA.	A Hybrid Artificial Neural Network (ANN) and Genetic Algorithm (GA) model for stock market forecasting.	The Hybrid GA-ANN model outperformed a single ANN, particularly during volatile periods like during the COVID-19 pandemic.
[23]	To predict stock market trends using both structured (historical data) and unstructured data (news, social media).	Big data approach using Spark MLlib with linear regression, random forest, decision trees, and sentiment analysis.	Linear regression, random forest, and naive Bayes models achieved high accuracy (80–98%) in stock market predictions based on big data.
[24]	To categorize stock market prediction methods and identify trends in machine learning research.	Systematic review of stock market prediction using machine learning, analyzing 122 research papers.	ANNs and SVMs were the most common algorithms for technical analysis, with ANNs outperforming decision trees and SVM in the Ghana Stock Exchange.
[25]	To develop a sustainable portfolio model to enhance returns and reduce risk during financial volatility triggered by the COVID-19 pandemic.	Regression models predicted stock prices using 2020 data, with sector-wise and company-level allocations based on curve-fitting and growth rate analysis. The model was tested on stock data from January to May 2021.	The proposed portfolio outperformed mutual funds during the evaluation period, demonstrating its effectiveness in managing financial uncertainty and achieving higher profits than traditional investment methods.
[26]	To validate the sustainability of a portfolio model designed to optimize stock market returns during and after the COVID-19 pandemic in India.	The study employed regression-based stock price predictions and sector-wise portfolio allocation, tested with 2020–2021 data, using a paired t-test to compare performance across periods.	The portfolio consistently outperformed NIFTY 50, SENSEX, and mutual funds, proving the methodology's effectiveness in enhancing returns during volatile market conditions.

3. Materials and Methods

3.1. Data

The database used for this study is composed of historical EU50 data, covering a period from 2 January 2015 to 13 November 2024, covering all working days in Europe over the last 9 years.

Our database contains 2514 rows and 14 variables, 7 of which were extracted from Investing (https://es.investing.com/, accessed on 14 November 2024) and contained data pertaining to the Date, Open, Close, High point, Low point, Volume or percentage change of the index. Another 6 variables were calculated using technical analysis indicators and the remaining variable was volatility. The calculations are detailed below.

Figure 1 shows the evolution of the closing price of a financial asset over time, with data covering the period from 2015 to 2024. Throughout this period, there is a general upward trend, albeit interspersed with episodes of significant falls, which is characteristic of financial market movements. In the early years, from 2015 to around 2018, the price experiences moderate fluctuations with a slight downward bias, reflecting a possibly unstable or corrective market. However, around 2019, a more marked recovery begins, possibly driven by a favorable economic environment or economic stimulus policies. The most notable point on the chart is the steep decline in 2020, which coincides with the global crisis caused by COVID-19. During this period, global markets experienced a sharp fall due to the uncertainty and economic impact of the pandemic. This fall is followed by a

rapid recovery, probably due to stimulus measures implemented by governments and central banks to stabilize economies. From 2021 onwards, the graph shows a sustained recovery and an upward trend. This could reflect the post-pandemic economic revival and the return of investor confidence. This positive trend continues until the end of the period in November 2024, albeit with some intermediate corrections that are characteristic of the dynamics of the financial markets.

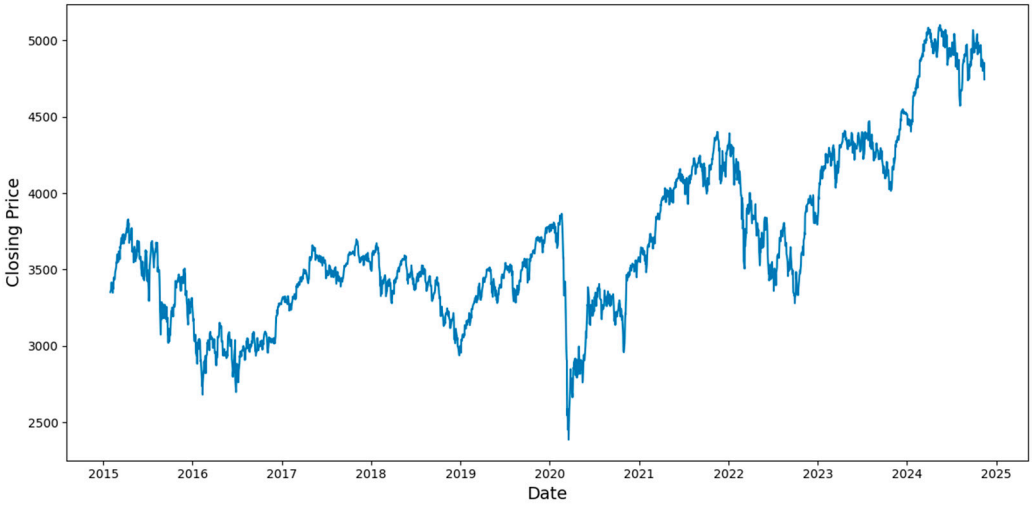


Figure 1. Evolution of closing price of EURO STOXX 50.

Table 2 provides key insights into the financial and technical behavior of the asset. The closing and opening prices exhibit close averages of 3674.11 and 3673.83, with standard deviations of 536.15 and 535.35, indicating considerable variability over time. The minimum and maximum values, at 2385.82 and 5100.90 for closing prices, and 2388.94 and 5096.86 for opening prices, highlight significant price fluctuations during the period. The high and low prices average at 3697.29 and 3649.59, respectively, with deviations of 534.67 and 536.92. Extreme values of 2461.57 for highs and 2302.84 for lows further reflect moments of heightened intraday volatility. The average trading volume stands at 40.28, with a deviation of 18.21, showing a wide range from 10.42 to 243.14, indicating varying market activity likely influenced by external factors or trend changes. The daily percentage change (% var) averages 0.000241 with a deviation of 0.01198, reflecting relatively small daily price shifts. Volatility averages at 0.0474, peaking at 0.2244, suggesting a generally stable asset with occasional spikes during uncertain or high-activity periods. The SMA and EMA indicators closely mirror price averages at 3667.68 and 3667.67, with deviations of 525.26 and 523.14, confirming their reliability in capturing central price trends. The MACD averages 4.93 with a deviation of 41.21, showing significant trend changes, with extremes from −283.78 to 103.38. The stochastic oscillator, with an average of 58.66 and a deviation of 28.83, oscillates between 0.61 and 99.46, capturing overbought and oversold conditions. Lastly, the RSI averages 53.85 with a deviation of 16.62, showing extreme values of 95.30 and 6.65, indicating moments of strong market pressure. The A/D indicator averages 2832.32 with a deviation of 1344.35, reflecting periods of significant accumulation and distribution, tied to trading volume and price movements. Together, these metrics underline the complexity of the asset's market behavior and its potential for significant price shifts.

Table 2. Descriptive analysis of variables.

	Average	Standard Deviation	Minimum	Median	Maximum
Closing	3674.11	536.15	2385.82	3546.83	5100.90
Opening	3673.83	535.35	2388.94	3545.08	5096.86
Maximum	3697.29	534.67	2461.57	3567.52	5121.71
Minimum	3649.59	536.92	2302.84	3526.69	5081.78
Volume	40.28	18.21	10.42	35.85	243.14
% var	0.000241	0.01198	−0.1240	0.0005	0.0924
Volatility	0.0474	0.0256	0.0135	0.0418	0.2244
SMA	3667.68	525.26	2658.35	3536.38	5041.61
EMA	3667.67	523.14	2797.98	3540.71	5033.23
MACD	4.93	41.21	−283.78	6.73	103.38
Stochastic	58.66	28.83	0.61	60.71	99.46
RSI	53.85	16.62	6.65	52.61	95.30
A/D	2832.32	1344.35	296.55	2942.06	4991.08

Figure 2 plots the evolution of volatility over time. At first glance, we see a trend of fluctuations in volatility, with peaks of different magnitudes, which is common in financial series where peaks tend to reflect events of high uncertainty or crisis. One of the highlights is the pronounced spike around 2020. This increase in volatility coincided with the COVID-19 crisis, a period in which global markets experienced high uncertainty. The sharp fall in financial asset prices and economic instability led to a sharp increase in volatility, reflected in the peak that stands out in the graph. After the peak in 2020, a decrease in volatility can be observed, although not in a completely stable manner. There are several subsequent moments, such as in 2022 and 2023, when volatility increases again, probably due to specific economic or geopolitical events that generated some uncertainty in the markets. Towards the end of the chart, around 2024, volatility appears to have stabilized at relatively low values compared to the peak in 2020, although it still exhibits some variability. This could indicate a recovery phase, in which markets have absorbed the effects of previous events and have reached a phase of greater stability. In conclusion, the chart illustrates how certain high-impact events, such as the COVID-19 pandemic, drastically affect volatility, generating significant spikes followed by periods of recovery or stabilization, albeit there are always fluctuations inherent to market dynamics.

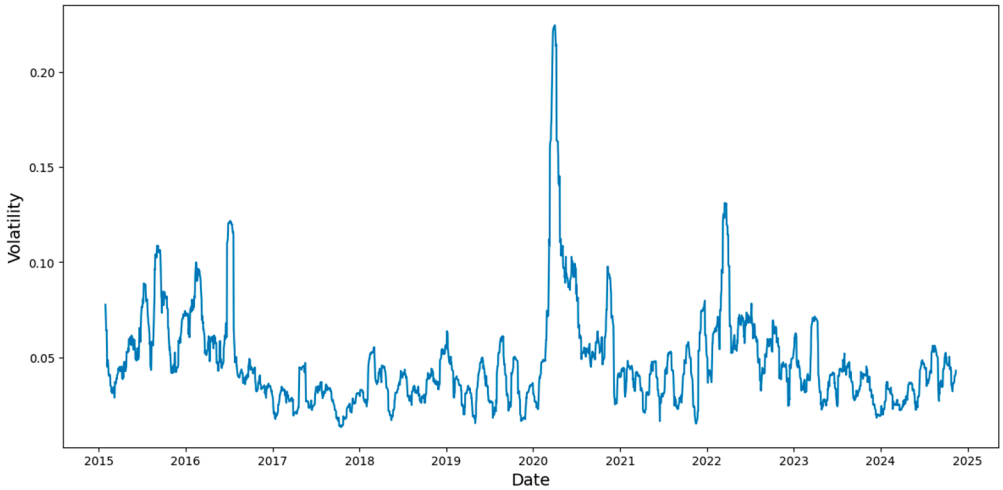


Figure 2. Evolution of volatility of EURO STOXX 50.

Figure 3 presents a candlestick chart, which shows the evolution of the price of a financial asset over a given period. In this type of chart, each candlestick represents the

opening, closing, high and low values reached during a specific day. Green candles indicate an increase in the price of the asset, i.e., the closing price is higher than the opening price, while red candles reflect a decrease, with a closing price lower than the opening price. Accompanying the candlesticks are two moving averages of different periods: a 20-day simple moving average (SMA), shown in blue, and a 100-day SMA, shown in orange. The 20-day SMA averages the closing prices of the last 20 days and, due to its shorter period, responds more quickly to changes in price, helping to identify short-term trends. On the other hand, the 100-day SMA, with a wider time window, is less reactive to recent fluctuations, providing insight into the long-term trend of the asset. In the chart, interesting behavior can be observed with regard to the relationship between the two moving averages. At the beginning of the period, the 20-day SMA is below the 100-day SMA, which is a sign of a downward trend where the price is progressively declining. In the middle of the chart, the 20-day SMA crosses above the 100-day SMA, which is a bullish crossover, indicating a possible trend reversal. However, towards the end of the chart, a bearish crossover is observed, as the 20-day SMA crosses back below the 100-day SMA, suggesting that the upward trend may be losing strength and that the price may resume a downward trajectory.



Figure 3. Candlestick chart for the last 100 days with the 20-day and 100-day SMA, green candles indicate an increase in the price of the asset and red candles indicate a decrease.

3.2. Method

3.2.1. Volatility

Volatility describes the change in the price of an asset over time. An asset is considered volatile when its price varies significantly and rapidly over a short period of time. We estimate the volatility indices following the logarithmic cash flow return method [27], where normality is assumed when calculating returns. The returns are found in our database as % var, so the volatility calculation used is as follows:

$$\hat{\sigma} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2} \quad (1)$$

where r_t is the daily return expressed as a percentage change for day t, $\bar{r}$ is the average return, T is the number of days and t is an index representing each individual day within the dataset.

This formula has been chosen to calculate volatility as the standard deviation of logarithmic returns, since it accurately measures the historical fluctuations of the asset, capturing recent trends through a 20-day moving window. Log returns are additive and eliminate biases in small changes, while annualization facilitates comparisons with market standards.

3.2.2. Technical Indicators

In this study, we utilize several technical indicators to enhance the analysis of financial markets by offering alternative interpretations of market activity. These indicators help to identify trends, momentum, and potential buy and sell signals. The selection and calculation of the technical indicators follow the guidelines provided by [28]. The key indicators used are as follows:

- Simple moving average (SMA): The SMA calculates the arithmetic mean of asset prices over a specified number of periods (n = 20). It smooths price data to identify trends by eliminating short-term fluctuations.
- Exponential Moving Average (EMA): The EMA assigns greater weight to more recent prices, making it more responsive to new data compared to the SMA. This indicator is useful for detecting recent changes in trends more quickly.
- Moving Average Convergence Divergence (MACD): The MACD is used to identify changes in the strength, direction, momentum, and duration of a trend. It is calculated by subtracting the 26-period EMA from the 12-period EMA. Buy and sell signals are often generated by observing the MACD line and its signal line (the 9-period EMA of the MACD).
- Stochastic Oscillator: This indicator compares the closing price of an asset to its price range over a specific period (n = 14), providing insight into the momentum of price changes. It is often used to detect overbought and oversold conditions in the market.
- Relative Strength Index (RSI): The RSI measures the speed and magnitude of price movements and helps to identify potential reversal points by determining whether an asset is overbought or oversold. The indicator oscillates between 0 and 100, with values above 70 generally indicating overbought conditions and values below 30 indicating oversold conditions.
- Accumulation/Distribution (A/D): This indicator assesses money flow in a security or market, based on the principle that large transactions gradually stabilize prices. Divergence between the indicator and prices may signal potential trend changes.

The integration of these technical indicators in this research is crucial, as it enhances the analysis of market dynamics by providing various perspectives on trends, momentum and price behavior. Indicators such as the SMA and EMA smooth price data to identify trends, while the stochastic oscillator and RSI provide momentum signals and detect overbought or oversold conditions. The MACD provides detailed information on the strength and duration of the trend, and the A/D indicator analyzes the money flow, signaling possible trend changes through divergences. These indicators support predictive models by providing structured quantitative data, allowing for more accurate forecasts of the market trend.

3.2.3. Neural Networks

For the predictive modeling component of this study, we employ neural network algorithms based on their demonstrated capabilities in financial forecasting. This approach leverages the known strengths of neural networks in capturing complex patterns and trends in stock market data. Specifically, we implement three types of neural networks: artificial neural networks (ANNs), convolutional neural networks (CNNs), and long

short-term memory networks (LSTMs), according to the methodologies from [29–31]. Each network is designed to capture different aspects of financial time series data.

- Artificial Neural Networks (ANNs): ANNs are well suited for financial data analysis due to their ability to model non-linear relationships. Our ANN model consists of multiple layers: an input layer with 64 neurons, followed by two hidden layers, each with 64 neurons. ReLU (Rectified Linear Unit) is used as the activation function in all layers to introduce non-linearity. The final output layer has one neuron, which predicts the closing price of the asset. The model is optimized using the Adam algorithm with a learning rate of 0.001, and the mean squared error (MSE) is employed as the loss function.
- Long Short-Term Memory Networks (LSTMs): LSTMs are a specific type of recurrent neural network (RNN) designed to capture long-term dependencies in time series data, which are often present in financial markets. Our LSTM model includes two LSTM layers, each with 64 units, which process the time series data sequentially. Dropout layers are also included to improve the model's generalization ability. This architecture allows the LSTM to maintain information across multiple time steps, making it ideal for capturing patterns in sequential financial data.
- Convolutional Neural Networks (CNNs): While CNNs are traditionally used for image processing, they can be adapted to analyze time series data by identifying local patterns within the data. Our CNN model is designed with a 1D convolutional layer (Conv1D) containing 32 filters and a kernel size of 3, followed by a MaxPooling layer to reduce the dimensionality. The flattened data are then passed through a fully connected (dense) layer with 50 neurons, and dropout is applied to mitigate overfitting. The output layer consists of a single neuron that is responsible for predicting asset prices. As with the other models, the Adam optimizer and MSE loss function are used to optimize performance.

3.2.4. Data Preprocessing and Training

Before training the models, all data were normalized using Min-Max scaling to ensure that the features fell within the range [0, 1], as required by neural networks. The data were split into training, validation and test sets to evaluate the model's ability to generalize to unseen data. In the case of LSTM and CNN models, data were reshaped into the appropriate three-dimensional format to meet the input requirements of these networks.

For training, we employed supervised learning methods, in which the model learns by comparing its predictions to the actual target values and adjusting its parameters to minimize errors. The Adam optimization algorithm was used for training, as it provides efficient convergence with a default learning rate of 0.001. Early stopping was implemented as a regularization technique to prevent overfitting by halting training once the model's performance on the validation set ceased to improve.

3.2.5. Evaluation Metrics

The performance of each model was evaluated using the following standard metrics, which were applied consistently across both the training and testing phases to ensure a comprehensive and unbiased assessment of the models' predictive capabilities:

- Root Mean Squared Error (RMSE): This metric measures the square root of the average squared differences between the predicted and actual values, providing a comprehensive measure of prediction accuracy.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \hat{x}_i)^2}{N}} \quad (2)$$

where i is a variable, N is the number of data points that were not lost, x_i is the time series of actual observations and $\hat{x}_i$ is the estimated time series.

- Mean Absolute Error (MAE): The MAE calculates the average magnitude of the errors between predicted and actual values, offering a straightforward measure of prediction accuracy that is less sensitive to outliers than RMSE.

$$MAE = \frac{\sum_{i=1}^{n}|y_i - x_i|}{n} \quad (3)$$

where y_i is the prediction, x_i is the actual value and n is the total number of data points.

- Coefficient of Determination (R^2): R^2 evaluates the proportion of variance in the dependent variable that can be predicted based on the independent variable(s). It ranges from 0 to 1, where 1 indicates perfect predictive power.

$$R^2 = \frac{\Sigma_i(y_i - f_i)^2}{\Sigma_i(y_i - \bar{y})^2} \quad (4)$$

where $\bar{y}$ is the mean of the observed data, $(y_i - f_i)^2$ is the sum of squares of the residuals of the dataset, and $\Sigma_i(y_i - \bar{y})^2$ is the total sum of squares (proportional to the variance of the data).

These metrics allow us to compare the performance of the different models and identify the most effective approach for forecasting asset prices.

3.2.6. Division of Data

To ensure an accurate evaluation of predictive model performance, it is crucial to properly divide the data into training, validation, and test sets. This process helps prevent overfitting and provides a realistic estimate of the model's performance on unseen data. Based on the work of [32], we observed that the appropriate partitioning of data should consider the problem's nature, data representativeness, time arrow and data redundancy. When working with time series data, the test data should follow the training set to avoid any temporal information leakage that could bias the results. The data were divided accordingly: 80% were used for training and validation of the models and the remaining 20% were used for testing.

The prediction model was developed using Python 3.12.

4. Results

Table 3 shows the comparative metrics for the performance of three prediction models—ANN, LSTM, and CNN—across various configurations, incorporating technical indicators to enhance model accuracy. CNN stands out for its precision, particularly when combining the closing price with indicators like EMA and SMA. For instance, CNN achieves an RMSE of 0.0301 and an R^2 of 0.9759 with SMA, and slightly better results with EMA, reaching an R^2 of 0.9771 in the test set. This suggests that CNN is highly effective at capturing local patterns in data and performs well with both trend and momentum indicators, including MACD and RSI. The A/D volume indicator also benefits CNN, enhancing its ability to leverage accumulation and distribution data without sacrificing accuracy.

The ANN model also demonstrates solid performance, though it is slightly less accurate than CNN. Using only the closing price, it achieves an R^2 of 0.9558 and benefits from trend indicators like EMA and RSI, which marginally improve its precision. For example, "CLOSING + EMA" yields an R^2 of 0.9510, providing a slight improvement over SMA. However, when momentum indicators such as MACD are added, ANN's performance remains stable, indicating that it adapts well to short-term trend variations. LSTM, while it specializes in capturing temporal sequences, achieves a comparatively modest performance, especially for momentum indicators like MACD. For instance, "CLOSING+MACD" yields an R^2 of 0.9339 on the test set, indicating that LSTM does not fully optimize this type of data, possibly due to additional noise introduced by MACD.

However, LSTM maintains a solid performance with smoothing indicators such as SMA and EMA, allowing it to better adapt to simpler data configurations.

In summary, CNN emerges as the most robust and precise model for this price prediction task, particularly when combining "CLOSING" with EMA and SMA. CNN is most effective at capturing local patterns and making optimal use of various technical indicators. While ANN also responds well to trend indicators, LSTM exhibits limitations with momentum indicators, suggesting it is better suited for less variable sequences. These results emphasize the importance of selecting indicators that align with each model's strengths, with CNN being the most versatile and accurate choice for price prediction in this configuration.

Table 3. Results of the performance of our models.

Model	Characteristics	Training RMSE	MAE	R^2	Test RMSE	MAE	R^2
ANN	CLOSING	0.0398	0.0278	0.9614	0.0376	0.0263	0.9642
	CLOSING+SMA	0.0442	0.031	0.9523	0.0417	0.0292	0.9559
	CLOSING+EMA	0.0432	0.0303	0.9544	0.0407	0.0283	0.9581
	CLOSING+MACD	0.0333	0.0232	0.973	0.0316	0.0225	0.9747
	CLOSING+EST	0.0308	0.0205	0.9769	0.0296	0.0212	0.9778
	CLOSING+RSI	0.0296	0.0202	0.9786	0.0289	0.021	0.9789
	CLOSING+A/D	0.0363	0.0249	0.9678	0.0346	0.0241	0.9697
LSTM	CLOSING	0.0468	0.0335	0.9466	0.0447	0.0335	0.9493
	CLOSING+SMA	0.0434	0.031	0.9539	0.0423	0.0312	0.9546
	CLOSING+EMA	0.0455	0.0318	0.9494	0.0436	0.0309	0.9518
	CLOSING+MACD	0.0475	0.034	0.9449	0.0454	0.0338	0.9478
	CLOSING+EST	0.0404	0.0279	0.9602	0.0392	0.0272	0.9611
	CLOSING+RSI	0.0445	0.031	0.9516	0.0422	0.0297	0.9548
	CLOSING+A/D	0.0462	0.033	0.9479	0.0452	0.0326	0.9482
CNN	CLOSING	0.0309	0.0218	0.9767	0.0295	0.0215	0.9779
	CLOSING+SMA	0.0284	0.0196	0.9804	0.0276	0.0194	0.9807
	CLOSING+EMA	0.0311	0.0216	0.9764	0.0298	0.0212	0.9776
	CLOSING+MACD	0.0319	0.0225	0.9752	0.0306	0.0219	0.9763
	CLOSING+EST	0.0278	0.0181	0.9812	0.0265	0.0183	0.9822
	CLOSING+RSI	0.027	0.0189	0.9822	0.0263	0.0186	0.9825
	CLOSING+A/D	0.0307	0.0215	0.977	0.0295	0.021	0.9779

Figure 4 shows a comparison of the actual EU50 values during the test period and the predictions obtained from the model with the best metrics of those compared, CNN+RSI. It shows that, in general, the predictions can accurately predict the trend in the actual data over time. However, at specific points, especially in the steepest terms, the predictions tend to underestimate or overestimate the actual values. Despite these discrepancies, the overall shape of the predictions closely matches actual price movements. Model performance metrics such as RMSE, MAE and R^2 indicate that the model accurately explains more than 98% of the variance in the test data. The small deviations observed in the graph may be due to natural market fluctuations that are difficult to predict accurately, especially when associated with highly volatile financial activities.

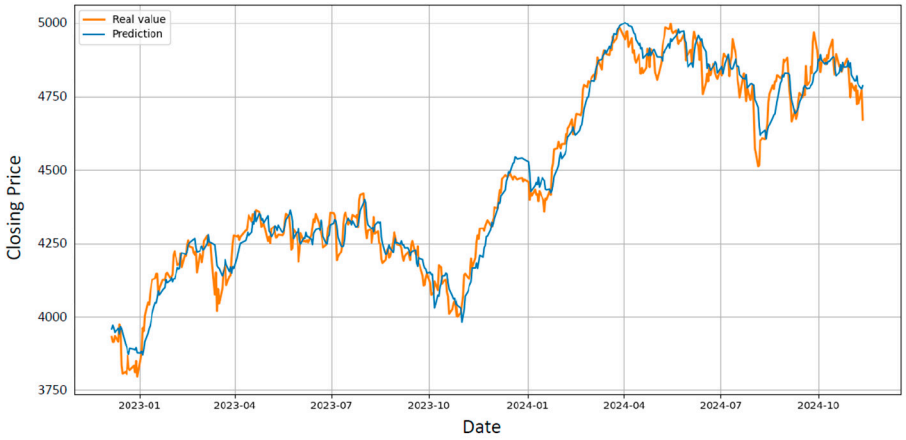


Figure 4. Comparison of predictions with actual model values with CNN+RSI characteristics.

5. Discussion

The results of this study emphasize the effectiveness of AI in enhancing financial decision-making, particularly through predictive modeling techniques like ANNs, LSTMs, and CNNs. Each model demonstrated specific strengths, with CNN achieving superior accuracy and robustness in most cases. This performance is particularly evident when incorporating technical indicators such as EMAs and SMAs. The architecture of CNN is well suited to identifying local patterns in financial time series data, such as intraday fluctuations and microtrends, making it a strong candidate for financial prediction tasks.

LSTM models, on the other hand, excel in capturing temporal dependencies, which are a critical factor for analyzing financial time series. Their sequential structure enables them to process historical data effectively, making them particularly adept at handling volatile market data. LSTM performed well with momentum-based indicators like MACD and RSI, which provide insights into market momentum and conditions of overbought or oversold assets. These results demonstrate the complementary nature of CNN and LSTM architectures, as they each leverage distinct strengths—spatial pattern recognition in CNN and temporal processing in LSTM.

However, this study has limitations. The scope of indicators used, while valuable, does not encompass external influences such as market sentiment derived from social media, economic news or geopolitical events. These factors play a significant role in stock price movements, and their integration could improve model accuracy. Another limitation is the temporal scope of the dataset, which reflects trends specific to a particular timeframe. Financial markets are influenced by evolving and complex factors, and extending the dataset to include various economic conditions and longer periods would enhance the generalizability of the models. This would allow for a more comprehensive understanding of how AI models perform across diverse economic scenarios.

While this study focused on stock price prediction as a baseline, its results suggest potential applications for other financial instruments such as currencies, commodities and derivatives. Extending the scope to include these instruments would evaluate the adaptability of the models to different asset classes. Additionally, exploring hybrid models that combine CNN's spatial pattern detection capabilities with LSTM's temporal processing strengths could mitigate overfitting and improve generalization. Such hybrid approaches may lead to more accurate and robust predictions across various financial contexts.

Lastly, as AI becomes integral to financial markets, relevant ethical considerations in terms of transparency, fairness, and accountability must be emphasized. These aspects are critical to building trust among stakeholders and preventing unintended consequences in automated decision-making processes. The development of interpretable models and robust validation processes will be essential to ensure that AI-driven financial tools are both reliable and equitable.

In summary, this study highlights the promise of AI-driven models, particularly CNN and LSTM, in financial prediction tasks. While the range of technical indicators and the temporal scope of the dataset present areas for improvement, addressing these challenges in future research can lead to more precise, resilient, and ethically responsible predictive models. Such advancements would support more informed and reliable financial decision-making.

6. Conclusions

This study demonstrates the transformative impact of AI on financial time series forecasting, highlighting the clear superiority of CNNs over both LSTM and traditional ANNs. Advanced AI architectures, such as CNNs, have shown remarkable generalization capabilities across different market scenarios, making them a powerful tool for financial prediction tasks. ANNs also exhibited robust and consistent performance, surpassing LSTMs in most configurations, especially when paired with well-selected technical indicators.

The results confirm that CNNs are particularly effective when it comes to identifying local patterns and microtrends in financial data. The use of technical indicators such as the Relative Strength Index (RSI) and Estimated Moving Averages (EsTs) significantly enhanced their predictive capability, making CNNs the most reliable model in this study. Meanwhile, ANNs demonstrated competitive performance, particularly when used in combination with RSI and EST, further solidifying their role as an effective alternative to CNNs.

LSTMs, while adept at capturing temporal dependencies, showed more variability in performance and were less robust in managing noisy financial data. Even with momentum-based indicators like MACD, their results did not reach the consistency observed for CNNs or ANNs. This suggests that LSTMs may require further optimization or alternative feature engineering strategies to fully harness their potential in financial forecasting.

This study highlights the importance of feature engineering in improving model performance. Indicators such as RSI and EST were especially effective at enhancing the models' predictive accuracy, while others, like SMA and EMA, showed limited impact in certain configurations. These findings underscore the necessity of systematically evaluating features to identify those that most effectively complement each model's architecture.

Despite these promising findings, certain limitations remain. Our analysis focused solely on technical indicators, excluding external factors such as market sentiment, economic news, or geopolitical events, which can significantly influence financial trends. Future research could address this by incorporating such variables, potentially leading to more holistic and accurate predictive models. Additionally, extending the study to other financial instruments, such as currencies or commodities, and exploring longer timeframes could enhance the generalizability of the results.

In conclusion, this study establishes CNNs as the most effective AI model for financial forecasting, particularly when supported by carefully chosen technical indicators like RSI and EST. ANNs also proved to be a reliable and competitive option, outperforming LSTMs in most scenarios. Future work should explore hybrid approaches that combine the spatial pattern recognition of CNNs with the temporal processing strengths of LSTMs, alongside integrating external factors for more comprehensive and reliable financial predictions. Such advancements would enable more informed and responsible decision-making in the financial sector.

Author Contributions: J.P.-D.: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, and visualization. L.S.-M.: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, and visualization. All authors have read and agreed to the published version of the manuscript.

Funding: This research is part of the International Chair Project on Trustworthy Artificial Intelligence and Demographic Challenge within the National Strategy for Artificial Intelligence (ENIA), in the framework of the European Recovery, Transformation and Resilience Plan. Reference: TSI-100933-2023-0001. This project is funded by the Secretary of State for Digitalization and Artificial Intelligence and by the European Union (Next Generation).

Data Availability Statement: The source of the data used is mentioned in Section 3.1 Data. Any additional questions may be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Turing, A. Computing Machinery and Intelligence. *Mind* **1950**, *LIX*, 433–460. https://doi.org/10.1093/mind/LIX.236.433.
2. Turing, A. Intelligent Machinery. In *Machine Intelligence 5*; Meltzer, D.M.B., Ed.; Edinburgh University Press: Edinburgh, UK, 1969; pp. 3–23.
3. Jain, A.; Sharma, A.; Sharma, D.; Singh, S. The Rise of AI: Trends, Challenges, and Emerging Solutions. *J. Artif. Intell.* **2023**, *12*, 102–118.
4. Dwivedi, Y.; Hughes, L.; Baabdullah, A.; Ribeiro-Navarrete, S.; Metri, P. AI and Its Societal Impact: A Modern Industrial Revolution? *J. Bus. Res.* **2021**, *131*, 552–560.
5. Gomber, P.; Kauffman, R.J.; Parker, C.; Weber, B.W. On the Fintech Revolution: Interpreting the Forces of Innovation, Disruption, and Transformation in Financial Services. *J. Manag. Inf. Syst.* **2018**, *35*, 220–265.
6. Aldasoro, I.; Gambacorta, L.; Korinek, A.; Shreeti, V.; Stein, M. *Intelligent Financial System: How AI Is Transforming Finance (No. 1194)*; Bank for International Settlements: Basel, Switzerland, 2024.
7. Alaka, H.A.; Oyedele, L.O.; Owolabi, H.A.; Kumar, V.; Ajayi, S.O.; Akinade, O.O.; Bilal, M. Systematic Review of Bankruptcy Prediction Models: Towards a Framework for Tool Selection. *Expert Syst. Appl.* **2018**, *94*, 164–184.
8. Pérez-Pons, M.E.; Parra-Dominguez, J.; Hernández, G.; Bichindaritz, I.; Corchado, J.M. OCI-CBR: A Hybrid Model for Decision Support in Preference-Aware Investment Scenarios. *Expert Syst. Appl.* **2023**, *211*, 118568.
9. Altman, E.I. Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy. *J. Financ.* **1968**, *23*, 589–609. https://doi.org/10.2307/2978933.
10. Altman, E.I.; Iwanicz-Drozdowska, M.; Laitinen, E.K.; Suvas, A. Financial Distress Prediction in an International Context: A Review and Empirical Analysis of Altman's Z-Score Model. *J. Int. Financ. Manag. Account.* **2017**, *28*, 131–171. https://doi.org/10.1111/jifm.12053.
11. Göçken, M.; Özçalıcı, M.; Boru, A.; Dosdoğru, A.T. Integrating Metaheuristics and Artificial Neural Networks for Improved Stock Price Prediction. *Appl. Soft Comput.* **2016**, *38*, 155–162.
12. Vijh, M.; Chandola, D.; Gautam, A.; Shukla, A. Stock Closing Price Prediction Using Machine Learning Techniques. *Procedia Comput. Sci.* **2020**, *167*, 599–606.
13. Bustos, O.L.; Pomares-Quimbaya, A. Stock Market Movement Forecast: A Systematic Review. *Expert Syst. Appl.* **2020**, *156*, 113464.
14. Bahrammirzaee, A. A Comparative Survey of Artificial Intelligence Applications in Finance: Artificial Neural Networks, Expert System and Hybrid Intelligent Systems. *Neural Comput. Appl.* **2010**, *19*, 1165–1195. https://doi.org/10.1007/s00521-010-0362-z.
15. Valle-Cruz, D.; Fernández-Sánchez, Y.; Ruvalcaba-Gómez, E.A.; Sandoval-Almazan, R. Social Media in Government: A Systematic Literature Review. *Transform. Gov. People Process Policy* **2022**, *16*, 243–267.
16. Nguyen, T.T.; Shirai, K.; Velcin, J. Sentiment Analysis on Social Media for Stock Movement Prediction. *Expert Syst. Appl.* **2015**, *42*, 9603–9611.
17. Uthayakumar, J.; Metawa, N.; Shankar, K.; Lakshmanaprabu, S.K. Financial crisis prediction model using ant colony optimization. *Int. J. Inf. Manag.* **2020**, *50*, 538–556. https://doi.org/10.1016/j.ijinfomgt.2018.12.001.
18. Yang, K.; Shi, Y.; Zhou, Y.; Yang, Z.; Fu, L.; Chen, W. Federated Machine Learning for Intelligent IoT via Reconfigurable Intelligent Surface. *IEEE Netw.* **2020**, *34*, 16–22. https://doi.org/10.1109/MNET.011.2000045.
19. Mhlanga, D. Industry 4.0 in Finance: The Impact of Artificial Intelligence (AI) on Digital Financial Inclusion. *Int. J. Financ. Stud.* **2020**, *8*, 45. https://doi.org/10.3390/ijfs8030045.
20. Chung, H.; Shin, K. Shik. Genetic algorithm-optimized multi-channel convolutional neural network for stock market prediction. *Neural Comput. Appl.* **2020**, *32*, 7897–7914. https://doi.org/10.1007/s00521-019-04236-3.
21. Ding, G.; Qin, L. Study on the prediction of stock price based on the associated network model of LSTM. *Int. J. Mach. Learn. Cybern.* **2020**, *11*, 1307–1317. https://doi.org/10.1007/s13042-019-01041-1.
22. Sharma, D.K.; Hota, H.S.; Brown, K.; Handa, R. Integration of genetic algorithm with artificial neural network for stock market forecasting. *Int. J. Syst. Assur. Eng. Manag.* **2022**, *13*, 828–841. https://doi.org/10.1007/s13198-021-01209-5.
23. Awan, M.J.; Rahim, M.S.M.; Nobanee, H.; Munawar, A.; Yasin, A.; Zain, A.M. Social Media and Stock Market Prediction: A Big Data Approach. *Comput. Mater. Contin.* **2021**, *67*, 2569–2583. https://doi.org/10.32604/cmc.2021.014253.
24. Nti, I.K.; Adekoya, A.F.; Weyori, B.A. A systematic review of fundamental and technical analysis of stock market predictions. *Artif. Intell. Rev.* **2020**, *53*, 3007–3057. https://doi.org/10.1007/s10462-019-09754-z.

25. Debnath, P.; Srivastava, H.M. Optimizing Stock Market Returns during Global Pandemic Using Regression in the Context of Indian Stock Market. *J. Risk Financ. Manag.* **2021**, *14*, 386. https://doi.org/10.3390/jrfm14080386.
26. Debnath, P.; Srivastava, H.M. Optimal Returns in Indian Stock Market during Global Pandemic: A Comparative Study. *J. Risk Financ. Manag.* **2021**, *14*, 592. https://doi.org/10.3390/jrfm14120592.
27. Pareja, M.; Ortuno, F.; Rojo, G. Risk Analysis and Financial Volatility: Theory and Applications in the IBEX 35. *Quant. Financ. Risk Manag.* **2019**, *45*, 210–223.
28. Picasso, A.; Merello, S.; Ma, Y.; Oneto, L.; Cambria, E. Technical Analysis and Sentiment Embeddings for Market Trend Prediction. *Expert Syst. Appl.* **2019**, *135*, 60–70. https://doi.org/10.1016/j.eswa.2019.06.014.
29. Aguilar, D.; González, F.; Alvarez, J. Machine Learning Algorithms for Financial Prediction: A Case Study on Stock Market Performance. *J. Financ. Technol. Res.* **2021**, *27*, 119–139.
30. Bosch Rué, E.; López, M.J.; Soria Olivas, E. Advances in LSTM and CNN for Time Series Forecasting: Applications to Financial Markets. *Appl. Intell.* **2019**, *49*, 989–1007.
31. Soria Olivas, E.; Bosch Rué, E.; Herrero, P. Deep Learning in Financial Markets: A Comparative Analysis of LSTM, CNN, and Hybrid Models. *Financ. Data Sci.* **2023**, *39*, 123–139.
32. Chollet, F. *Deep Learning with Python*; Manning Publications, 2018.

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